

Finite choice at measurable strength

Prikry deletion, cyclic label obstructions, and profinite phases

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Abstract. Let $M \models \text{ZFC}$, let $U \in M$ be a normal measure on κ , and let $V = M[c]$ be its ordinary Prikry extension. Put $q = [\text{ran}(c)]$ modulo finite symmetric difference. We prove that no nonempty finite family of selectors on the cofinal ω -quotient Q_κ is ordinal definable in V from q and ground-model parameters. More generally, for a finite permutation group $\Gamma \leq S_m$ acting on a nonempty ground-model set Y , a definable admissible labeling rule exists exactly when every $g \in \Gamma$ fixes some element of Y . Allowing a nonempty finite definable family of rules does not weaken this criterion.

The necessity proof uses a single-point deletion of the generic sequence, a finite permutation root, and a finite table of responses. Thus it does not require an ultraextending cardinal or an elementary self-embedding. A second result shows that every nonempty definable family of ultrafilters on the *set of representatives* q has a phase image invariant under translation by 1 in $\widehat{\mathbb{Z}}$, hence dense. A compact such family has at least $2^{2^{\aleph_0}}$ members, and this bound is attained. An explicit countably infinite definable family realizes a single dense integer orbit, making the local finite obstruction sharp. We also give sharp finite-additivity comparisons and combine the new obstructions with the existing normal-trace construction to obtain an exact measurable-strength consistency package.

Main advance relative to the supplied archive

The Prikry-extension part of the last open-question group in the supplied *Synthesis* has an affirmative answer: the finite-selector obstruction and the necessity of its cyclic label criterion already hold after Prikry forcing at one measurable cardinal. The finite-family label theorem and the profinite-density conclusion strengthen that answer. A countable ultrafilter construction also gives the exact least size, $\aleph_0$, of a nonempty definable family of ultrafilters on the generic class. This is *not* a proof that rigidity alone implies these properties in an arbitrary universe. Nor does it establish an inconsistency of ultraextendingness, I0 , or any other large-cardinal axiom.

Proof status. The report supplies conventional proofs in English. Classical Prikry forcing facts are identified as inputs; the new deductions are proved below. The supplied Lean files were used as reference, not as a verification of these forcing arguments. The additional Python checks concern only finite permutation identities. Novelty is assessed relative to the supplied reports; an exhaustive publication-priority claim is not made.

Contents

1	What is new, and what is being continued	2
2	The setting and the classical forcing inputs	3
3	The deletion principle for ground-valued observations	5
4	Encoding an arbitrary finite permutation	5
5	The finite-family cyclic obstruction	6
6	The converse: cyclic stabilizers of incidence words	8
7	Profinite phases of ultrafilters	10
8	Charges: the exact finite-additivity contrast	14
9	The ground trace, rigidity, and the canonical core	16
10	The precise consistency calibration	19
11	Consequences for the research program	21
A	Finite checks and a proof audit	21
B	Source and notation concordance	22

1 What is new, and what is being continued

The supplied archive contains the research plan, a synthesis of twenty-seven continuations, and Lean reference material. The immediate source for this report is [1], particularly its sections on finite selectors, cyclic stabilizers, internal normal measures, and exact consistency strength. The final open-question group asks whether the finite-label necessity theorem and the no-finite-family-of-selectors theorem hold in a Prikry extension of a measurable cardinal. Those are the questions answered here.

The same sentence in the synthesis also asks whether the conclusions follow from rigidity alone. The two formulations should be separated. The assertion that a particular forcing construction has a property is weaker than the assertion that every instance of an abstract rigidity principle has it. Our proof uses a feature of a generic presentation that is not part of the definition of rigidity: a sufficiently late deletion preserves both the genericity and the truth of a previously decided ground-valued observation.

Topic	Already in the synthesis	Contribution here
Finite selectors	No finite definable family at an ultraextending cardinal	The same obstruction in ordinary Prikry extensions, even with the generic class and arbitrary ground parameters
Cyclic label criterion	Single-rule classification for finite label sets	Finite-family necessity in Prikry extensions; ground label sets may be infinite
Ultrafilter-valued choice	Finite-family obstruction at ultraextending cardinals	Dense phase images in $\widehat{\mathbb{Z}}$; exact local thresholds $\aleph_0$ and, under compactness, $2^{\aleph_0}$
Rigidity and tail measures	Measurable-strength calibration and Prikry cores	A full compatibility proof incorporating all finite cyclic obstructions at the same strength
Positive label rules	Incidence words with cyclic stabilizers	Reproduced with details, not claimed as a new method

The broader motivation remains the analysis of exacting and ultraextending cardinals in [4, 5]. Their embedding arguments explain why these quotient obstructions arose in the archive. The present proof isolates a forcing mechanism sufficient for them. The use of finite cycles in quotient choice arguments also has precedent in the discussion recorded in [6, Remark 2.4]; the result here is the Prikry transfer and its finite-family amplification, not the observation that a cycle can obstruct a choice of one of its elements.

Reading guide

The main new proof is in Sections 3–5. Section 6 proves the converse and gives the complete classification. Sections 7–8 contain the ultrafilter and probability refinements. Sections 9 and 10 give the inner-model comparison and the precise consistency statement.

2 The setting and the classical forcing inputs

2.1 Models, quotients, and parameters

Throughout Sections 2–9, M is a transitive model of ZFC, $U \in M$ is a normal nonprincipal κ -complete ultrafilter on κ as computed in M , and $c = \langle c_n : n < \omega \rangle$ is increasing and Prikry generic for U over M . We write

$$V = M[c], \quad a_c = \text{ran}(c), \quad q = [a_c].$$

The same arguments may be read as forcing arguments over a ground universe. The metamathematical consistency statement at the end does not assume a transitive model exists.

For an uncountable ordinal λ with $\text{cf}(\lambda) = \omega$, define

$$D_\lambda = \{a \subseteq \lambda : \sup a = \lambda \text{ and } \text{ot}(a) = \omega\}, \quad a =^* b \iff |a \triangle b| < \omega, \quad Q_\lambda = D_\lambda / =^*.$$

All these sets, and all unqualified cardinalities below, are computed in V . A representative and its increasing enumeration will be distinguished where indices matter. For $q \in Q_\lambda$ and $A \subseteq \lambda$, put

$$\text{Tail}(q, A) \iff a \setminus A \text{ is finite for some } a \in q.$$

“Some” can equivalently be replaced by “every.”

Definition 2.1 (Parameter convention). The notation $\text{OD}_{M \cup \{q\}}^V$ means the sets uniquely definable in V from q , finitely many parameters belonging to M , and finitely many ordinals. It does *not* allow the class M as a predicate in the definition. Similarly, $\text{OD}_{V_\kappa \cup \{q\}}^V$ allows finitely many elements of V_κ and the single parameter q . In particular, q is allowed but its representatives are not automatically allowed.

A finite family $\mathcal{F}$ being definable does not say that a member of $\mathcal{F}$ is definable. Our main theorem is deliberately stated for families and does not make that invalid reduction.

2.2 The forcing and its standard properties

A condition of $\mathbb{P}_U$ is a pair (s, A) , where s is a finite increasing sequence in κ , $A \in U$, and every element of A is above the stem s . Stronger conditions extend the stem by points from A and shrink A . We use the convention that a smaller condition is stronger.

Input A (Classical Prikry facts). Normal Prikry forcing preserves cardinals, adds no bounded subset of κ , and changes $\text{cf}(\kappa)$ to ω . An increasing cofinal ω -sequence d in κ is generic over M precisely when it is almost contained in every member of U . For such d , its generic filter is

$$G_d = \{(s, A) : s \text{ is an initial segment of } d \text{ and } \text{ran}(d) \setminus \text{ran}(s) \subseteq A\}.$$

The characterization is Mathias’ criterion. The input is classical; see [3] and the modern proofs in [2, Section 3, especially Corollary 3.15 and Theorem 3.17]. We use the forcing theorem and truth lemma in their usual set-forcing form. Everything below the stated input is proved here, except where an earlier archive construction is explicitly credited.

Lemma 2.2 (No new low-rank sets). $V_\kappa^V = V_\kappa^M$. Consequently $\text{OD}_{V_\kappa \cup \{q\}}^V \subseteq \text{OD}_{M \cup \{q\}}^V$; the two models have the same reals and the same sets of reals. Moreover, κ is an uncountable strong limit cardinal of cofinality ω in V .

Proof. The cardinal κ is strongly inaccessible in M . We prove equality of V_α for $\alpha < \kappa$ by induction. The limit step is immediate. At a successor step, a ground bijection codes V_α^M by an ordinal below κ , since $|V_\alpha^M|^M < \kappa$. A new subset of this set would therefore code a new bounded subset of κ , contrary to Input A. This proves the induction and the equality. Reals, sets of reals, and their standard finite-rank codes all lie below V_κ . Finally cardinals are preserved, and every power set of an ordinal below κ is unchanged and has cardinality below κ . Thus κ remains strong limit, while its cofinality becomes ω . $\square$

Lemma 2.3 (Finite changes preserve the presentation). If d is the increasing enumeration of any member of q , then d is Prikry generic over M and $M[d] = M[c]$. In particular, deleting c_k from c gives such a d . If $(s, A) \in G_c$ and $k \geq |s|$, then $(s, A) \in G_d$ for this deletion.

Proof. A finite change preserves cofinality and eventual membership in each $A \in U$, so genericity follows from Mathias’ criterion. The two sequences are interdefinable from their finite symmetric difference and finitely many ordinals. All those additional parameters belong to M . Hence the two extensions contain one another. For the last assertion the deletion does not change the initial stem s and does not add a point outside A ; the displayed description of G_d applies. $\square$

3 The deletion principle for ground-valued observations

The following lemma is the central forcing tool. It does not refer to an elementary embedding and does not assert that V has a nontrivial automorphism.

Lemma 3.1 (A late deletion preserves a decided observation). Let $A \in M$. Suppose Φ is a fixed formula that, from an increasing generic sequence e , its class $[\text{ran}(e)]$, ground parameters, and ordinal parameters, uniquely specifies an element $h(e) \in A$ in the generic extension. It is enough to assume this uniqueness in the presentations under consideration, including c and its finite changes. Then there is $k_0 < \omega$ such that for every $k \geq k_0$, if d is obtained from c by deleting its k th term, then

$$h(d) = h(c).$$

In particular, if a fixed ground function $T : A \rightarrow A$ satisfies $h(d) = T(h(c))$ for every sufficiently late deletion, then $T(h(c)) = h(c)$.

Proof. Write $a = h(c)$. Since $a \in A$ and $A \in M$ with M transitive, $a \in M$. By the truth lemma there is a condition $(s, B) \in G_c$ forcing that the unique observation, formed using the canonical generic sequence and its class, is the ground object $\check{a}$. This statement includes whatever uniqueness assertion is needed for Φ .

Take $k_0 = |s|$. If $k \geq k_0$ and d deletes c_k , Lemma 2.3 shows that G_d contains (s, B) and that $M[d] = V$. Therefore the very same forced definition, interpreted using d , has value a . This proves $h(d) = a = h(c)$. The concluding fixed-point identity follows by comparing the two formulas for $h(d)$. $\square$

Remark 3.2 (What the lemma does and does not fix). The object $h(e)$ is allowed to depend on the actual enumeration e ; otherwise the lemma would have little content. The definable family from which the observation is extracted will depend only on ground parameters and the unchanged class q , so that family is the same set in $M[c]$ and $M[d]$. The observation is low-complexity enough to be a ground object, although its inputs and the members of the family may be new and have high rank. We compare two generic presentations of one extension, not images under an automorphism of that extension.

A useful example is a finite response table with entries in a ground set Y . Every such table is in M , because M is closed under forming finite sets and ordered tuples of its elements. A finite set of these tables is also in M . No definable enumeration of that finite set is required. A second example, used in Section 7, is an arbitrary subset of a fixed real-coded compact space: this is a ground object by Lemma 2.2.

4 Encoding an arbitrary finite permutation

Lemma 4.1 (One deletion realizes one permutation). Let E be a nonempty finite set, let $\tau \in \text{Sym}(E)$, and fix an injective tag map $t : E \rightarrow N$ for some positive integer N . There are representatives $a_e(c) \in D_\kappa$, for $e \in E$, whose classes are pairwise distinct, such that for every sufficiently late deletion d of c ,

$$a_e(d) =^* a_{\tau(e)}(c) \quad (e \in E). \quad (4.1)$$

They can be defined uniformly from c, τ, t by

$$a_e(c) = \{c_n + t(\tau^{-n}(e)) : n < \omega\}. \quad (4.2)$$

Proof. Normality implies that U contains every ground club in κ . In particular c_n is eventually a nonzero limit ordinal. For completeness, the club assertion follows because on a measure-one subset of the complement of a club C , the regressive map $\alpha \mapsto \sup(C \cap \alpha)$ would have to be constant. A constant value would bound that subset below the next point of C , contradicting nonprincipality and κ -completeness.

For all sufficiently large n , the finite block $\{c_n + i : i < N\}$ is below c_{n+1} , since c_{n+1} is a limit ordinal. The blocks are therefore eventually disjoint and in increasing order. Formula (4.2) chooses one point in each block. It defines a cofinal set of order type ω ; any exceptional early blocks contribute only finitely many points. For $e \neq f$, injectivity of t and of τ^{-n} gives distinct points in every sufficiently late block. Thus $a_e(c) \cap a_f(c)$ is finite, so their classes are distinct.

Suppose d deletes c_k . For $n \geq k$, $d_n = c_{n+1}$, and

$$d_n + t(\tau^{-n}(e)) = c_{n+1} + t(\tau^{-(n+1)}(\tau(e))).$$

This is the $(n+1)$ st term of the defining sequence for $a_{\tau(e)}(c)$. The initial discrepancies are finite, proving (4.1). $\square$

Lemma 4.2 (A finite permutation root). Let $g \in S_m$ and $L \geq 1$. On $E = m \times L$ define

$$\tau(i, a) = \begin{cases} (i, a+1), & a+1 < L, \\ (g(i), 0), & a = L-1. \end{cases} \quad (4.3)$$

Then τ is a permutation and $\tau^L(i, 0) = (g(i), 0)$ for each $i < m$.

Proof. The inverse sends (i, a) with $a > 0$ to $(i, a-1)$ and sends $(i, 0)$ to $(g^{-1}(i), L-1)$. Starting at $(i, 0)$, the first $L-1$ steps visit the remaining levels over i , and the L th step reaches $(g(i), 0)$. This includes $L=1$. $\square$

The root is the amplification device. A permutation of a family of size h need not fix each member, but its L th power does when $L = \text{lcm}(1, \dots, h)$. We arrange that this L th power still acts as the desired permutation g on a distinguished input tuple.

5 The finite-family cyclic obstruction

Fix $1 \leq m < \omega$; the case $m = 0$ has a trivial group and requires no argument. Let $D_m(\kappa)$ consist of ordered m -tuples $\vec{a} = (a_i)_{i < m}$ from D_κ whose finite-difference classes are pairwise distinct. If $\Gamma \leq S_m$, its action is

$$(g \cdot \vec{a})_i = a_{g^{-1}(i)}.$$

Let $Y \in M$ be nonempty and let a left action of Γ on Y belong to M . The set Y need not be finite.

Definition 5.1 (Admissibility and the cyclic condition). A rule $f : D_m(\kappa) \rightarrow Y$ is *admissible* if it is invariant under finite changes in each coordinate and satisfies $f(g \cdot \vec{a}) = g \cdot f(\vec{a})$ for every $g \in \Gamma$. Write

$$\text{CF}(\Gamma, Y) \iff (\forall g \in \Gamma)(\exists y \in Y) g \cdot y = y.$$

Equivalently, every cyclic subgroup of Γ has a fixed point in Y . The label y may depend on the cyclic subgroup.

Theorem 5.2 (Finite-family necessity in a Prikry extension). If a nonempty finite family $\mathcal{F}$ of admissible rules belongs to $\text{OD}_{M \cup \{q\}}^V$, then $\text{CF}(\Gamma, Y)$ holds.

Proof. Let $h = |\mathcal{F}| \geq 1$ and $L = \text{lcm}(1, \dots, h)$. Fix $g \in \Gamma$. We will obtain a fixed point of g on Y .

Step 1: enlarge the finite test configuration. Use Lemma 4.2 to define $E = m \times L$ and $\tau \in \text{Sym}(E)$. Choose distinct natural-number tags for E , and use Lemma 4.1 to obtain pairwise inequivalent representatives $a_e(c)$ for $e \in E$. Let

$$I = \text{Inj}(m, E), \quad u_0(i) = (i, 0).$$

The set I is finite and belongs to M .

Step 2: replace each possibly high-rank rule by a finite table. For $f \in \mathcal{F}$, define its response table

$$T_f^c : I \longrightarrow Y, \quad T_f^c(u) = f((a_{u(i)}(c))_{i < m}),$$

and put

$$H(c) = \{T_f^c : f \in \mathcal{F}\}.$$

The tables may identify different rules, but $1 \leq |H(c)| \leq h$. Every table is a finite tuple of ground objects, so $H(c) \in M$. More uniformly, it belongs to the ground set $[Y^I]^{\leq h}$. The observation $H(c)$ is defined using c and a fixed definition of $\mathcal{F}$ from q , ground parameters, and ordinals.

Step 3: one deletion permutes the table set. On Y^I , define the ground permutation

$$(P_\tau T)(u) = T(\tau \circ u).$$

Delete a sufficiently late term from c to obtain d . The family $\mathcal{F}$ is unchanged, because $M[d] = M[c]$, the ground parameters are unchanged, and $[\text{ran}(d)] = q$. By (4.1) and finite-modification invariance,

$$T_f^d(u) = T_f^c(\tau \circ u).$$

Consequently $H(d) = P_\tau[H(c)]$. Lemma 3.1, applied to this ground-valued observation, gives

$$H(c) = P_\tau[H(c)]. \quad (5.1)$$

This equality is an equality of *sets of tables*; it does not presuppose that a particular rule or table has been definably selected.

Step 4: take a power that fixes every table. Equation (5.1) says that P_τ induces a permutation of the nonempty finite set $H(c)$. Each of its cycles has length at most h , hence its length divides L . Therefore, for every $T \in H(c)$,

$$P_\tau^L T = T, \quad T(\tau^L \circ u) = T(u) \quad (u \in I).$$

Choose any $T = T_f^c \in H(c)$. This is an ordinary existential choice in the proof, not a definable selection. Applying the last identity at u_0 and using $\tau^L \circ u_0 = u_0 \circ g$ yields

$$T(u_0) = T(u_0 \circ g).$$

The tuple with coordinates $a_{u_0(g(i))}(c)$ is $g^{-1} \cdot (a_{u_0(i)}(c))_{i < m}$. Equivariance therefore gives

$$T(u_0 \circ g) = g^{-1} \cdot T(u_0).$$

Thus $T(u_0)$ is fixed by g . Since $g \in \Gamma$ was arbitrary, $\text{CF}(\Gamma, Y)$ follows. $\square$

Remark 5.3 (Why two tempting shortcuts fail). One cannot simply select the least member of $\mathcal{F}$: its members need not be ordinal definable, and no suitable definable well-order of the new rules has been given. Nor is it enough to repeat a singleton argument for one chosen member: deletion may move that member's table to another one. The finite response set and the L th-root construction are precisely what eliminate both gaps.

5.1 Selectors and related structures

For $1 \leq r < n < \omega$, a selector of type (n, r) on Q_κ is a function

$$s : [Q_\kappa]^n \longrightarrow [Q_\kappa]^r, \quad s(B) \subseteq B.$$

Corollary 5.4 (No finite definable family of selectors). For every $1 \leq r < n < \omega$, there is no nonempty finite family of (n, r) -selectors on Q_κ in $\text{OD}_{M \cup \{q\}}^V$. In particular this holds in $\text{OD}_{V_\kappa \cup \{q\}}^V$ and in $\text{OD}_{V_\kappa}^V$.

Proof. To a selector s associate

$$f_s(\vec{a}) = \{i < n : [a_i] \in s(\{[a_j] : j < n\})\} \in [n]^r.$$

This is invariant under finite modifications and is equivariant for the natural S_n -action on $[n]^r$. A nonempty finite definable family of selectors would give a nonempty finite definable family of these rules. But an n -cycle acts transitively on n , so its only invariant subsets are $\emptyset$ and n . It fixes no member of $[n]^r$ when $0 < r < n$. Theorem 5.2 gives a contradiction. The parameter inclusions follow from Lemma 2.2. $\square$

Corollary 5.5. There is no nonempty finite $\text{OD}_{M \cup \{q\}}^V$ family of linear orders on Q_κ , and none of tournaments on Q_κ . There is no $\text{OD}_{M \cup \{q\}}^V$ injection from Q_κ into the ordinals.

Proof. A linear order chooses the smaller of two distinct elements. A tournament chooses the source of the unique directed edge between them. These conversions send a finite definable family to a nonempty finite definable family of binary selectors. An injection into the ordinals similarly chooses the element with smaller image. Apply Corollary 5.4 with $(n, r) = (2, 1)$. $\square$

The obstruction is definability, not Choice

The ambient extension V satisfies ZFC. It has selectors and well-orders of Q_κ . The conclusion is that these cannot be captured by a nonempty finite family defined from the specified ground parameters and q . Adding an individual generic representative as a parameter would invalidate the deletion argument; it is not an allowed substitution for q .

6 The converse: cyclic stabilizers of incidence words

This section expands the positive construction already present in the synthesis [1, Section 12.6]. The finite-family necessity theorem above is new relative to that source; the incidence-word method is not. We include the proof to make the resulting classification self-contained.

6.1 Shift-tail equivalence

Let $\vec{a} \in D_m(\kappa)$. The union $\bigcup_{i < m} a_i$ has order type ω : below any ordinal smaller than κ , it has only finitely many points. Enumerate it increasingly as $\langle \delta_n : n < \omega \rangle$ and form the incidence word

$$w_{\vec{a}}(n) = \{i < m : \delta_n \in a_i\} \in \mathcal{P}(m) \setminus \{\emptyset\}.$$

Each label i appears infinitely often. Each pair of distinct labels is separated infinitely often, because otherwise the corresponding representatives would have finite symmetric difference.

For two such words, write $w \sim v$ when there are $r, s < \omega$ such that

$$w(r + n) = v(s + n) \quad (n < \omega).$$

This is an equivalence relation. Changing finitely many points of the coordinates changes the incidence word only within its $\sim$ -class: beyond the largest changed ordinal the two unions, and their incidence labels, agree, possibly with different starting indices. Also $w_{g \cdot \vec{a}} = g \cdot w_{\vec{a}}$, where g acts on subsets of m by image.

Lemma 6.1 (The stabilizer is cyclic). Let w be an incidence word in which any two labels are separated infinitely often. Then the stabilizer of $[w]_{\sim}$ in Γ is cyclic.

Proof. Define

$$G_w = \{(d, g) \in \mathbb{Z} \times \Gamma : w(n + d) = g \cdot w(n) \text{ for all sufficiently large } n\}.$$

Negative values of d cause no problem because n is taken sufficiently large. The identity belongs to G_w . Composing the eventual identities for (d, g) and (e, h) gives the identity for $(d + e, gh)$, and reversing one gives the inverse identity. Thus G_w is a subgroup.

The projection $G_w \rightarrow \mathbb{Z}$ is injective. Indeed, if $(0, g) \in G_w$ and g moves i , then for all sufficiently large n ,

$$i \in w(n) \iff g^{-1}(i) \in w(n),$$

contradicting infinite separation of those two labels. Hence the kernel is trivial. A subgroup of $\mathbb{Z}$ is cyclic: a nonzero subgroup is generated by its least positive element, and the zero subgroup is trivial. Therefore G_w is cyclic. Its image in Γ is exactly the stabilizer of $[w]_{\sim}$, since an equality of two shifted tails is precisely an eventual equality with an integer shift. An image of a cyclic group is cyclic, proving the claim. $\square$

Theorem 6.2 (Complete finite-family classification). For a finite $\Gamma \leq S_m$ and a nonempty ground Γ -set Y , the following are equivalent in $V = M[c]$:

1. $\text{CF}(\Gamma, Y)$ holds.
2. An admissible rule $D_m(\kappa) \rightarrow Y$ belongs to $\text{OD}_{M \cup \{q\}}^V$.
3. A nonempty finite family of admissible rules belongs to $\text{OD}_{M \cup \{q\}}^V$.

For finite label sets and their action coded by natural numbers, the same equivalence holds with $\text{OD}_{V_\kappa \cup \{q\}}^V$ in place of $\text{OD}_{M \cup \{q\}}^V$. In the positive case a rule already belongs to $\text{OD}_{V_\kappa}^V$, without using q .

Proof. The implication (2) $\Rightarrow$ (3) uses the singleton family. Theorem 5.2 proves (3) $\Rightarrow$ (1). For (1) $\Rightarrow$ (2), fix in M a well-order $<$ of the reals and a well-order $\triangleleft$ of Y . All finite-alphabet words can be coded by reals in a fixed way. There are no new reals, so $<$ orders all the words under consideration in V as well.

For each Γ -orbit $\mathcal{O}$ of $\sim$ -classes of valid words, let $v_{\mathcal{O}}$ be the $<$ -least word whose class belongs to that orbit. Let $H_{\mathcal{O}}$ be the stabilizer of $[v_{\mathcal{O}}]_{\sim}$. It is cyclic by Lemma 6.1. A generator has a fixed point by $\text{CF}(\Gamma, Y)$; that point is fixed by the whole stabilizer. In the trivial-stabilizer case every label works. Let $y_{\mathcal{O}}$ be the $\triangleleft$ -least label fixed by $H_{\mathcal{O}}$.

For $[w]_{\sim} = g \cdot [v_{\mathcal{O}}]_{\sim}$, set

$$F([w]_{\sim}) = g \cdot y_{\mathcal{O}}.$$

If h is another such choice, then $h^{-1}g \in H_O$, so the two values agree. This defines an equivariant function of the shift-tail class. Composing it with $\vec{a} \mapsto [w_{\vec{a}}]_{\sim}$ produces an admissible rule. It is definable from $<, \triangleleft, Y$, and the action, all ground parameters; it does not use c or q .

When Y and the action are finite codes, these parameters apart from $<$ have finite rank, and a relation well-ordering the reals has rank below $\omega + 10$. Thus they all belong to V_κ , giving the stronger positive assertion and the corresponding equivalence for $\text{OD}_{V_\kappa \cup \{q\}}^V$. $\square$

Remark 6.3 (Elementwise, not simultaneous, fixed points). The criterion cannot be strengthened to “ Γ has a common fixed label.” For example let $\Gamma = S_3$ and take Y to be the disjoint union of its natural three-point set and its two-point sign-action set. A transposition fixes a point in the first component; a 3-cycle fixes both points in the second. Thus every group element fixes something, but no label is fixed by the whole group. The positive construction still works. This example and the underlying cyclic-stabilizer explanation already appear in the archive. They are useful safeguards against asking more of the necessity proof than can be true.

7 Profinite phases of ultrafilters

The obstruction for ultrafilters admits a structural strengthening that is not merely another finite-cycle contradiction. The target is the profinite completion of the integers,

$$\widehat{\mathbb{Z}} = \varprojlim_{n|m} \mathbb{Z}/n\mathbb{Z} = \{(x_n)_{n \geq 1} : x_m \bmod n = x_n \text{ whenever } n \mid m\}.$$

We use its compact Hausdorff topology inherited from the product of the finite discrete spaces $\mathbb{Z}/n\mathbb{Z}$.

7.1 The index cocycle and its residues

For $a, b \in q$, define an integer

$$\chi_a(b) = |b \setminus a| - |a \setminus b|.$$

For any $a, a', b \in q$, finite cancellation gives

$$\chi_{a'}(b) = \chi_a(b) - \chi_a(a'). \quad (7.1)$$

One way to verify this is to take a common finite set outside which all three representatives agree. The displayed quantities then become differences of the three finite cardinalities inside that set. If $a' = a \setminus \{\xi\}$, then $\chi_a(a') = -1$, so

$$\chi_{a'}(b) = \chi_a(b) + 1. \quad (7.2)$$

For $n \geq 1$ and $r \in \mathbb{Z}/n\mathbb{Z}$, define the finite partition

$$E_{n,r}^a = \{b \in q : \chi_a(b) \equiv r \pmod{n}\}.$$

Here an ultrafilter on q means an ultrafilter on the *full ambient Boolean algebra* $\mathcal{P}(q)^V$, principal or nonprincipal. It is not the internal tail measure on κ discussed later.

Every ultrafilter W on q contains exactly one cell of the n th partition. Call its residue $r_n^a(W)$. The choices are compatible under divisibility, since the cell at a finer modulus is contained in its reduction at a coarser one. Hence

$$t_a(W) = (r_n^a(W))_{n \geq 1} \in \widehat{\mathbb{Z}}.$$

Equation (7.2) implies

$$t_{a'}(W) = t_a(W) + 1 \quad \text{if } a' = a \setminus \{\xi\}. \quad (7.3)$$

Theorem 7.1 (Profinite-density theorem). Let $\mathcal{W} \in \text{OD}_{M \cup \{q\}}^V$ be a nonempty family of ultrafilters on q . Its size is not assumed finite. The phase image

$$H_a = \{t_a(W) : W \in \mathcal{W}\} \subseteq \widehat{\mathbb{Z}}, \quad a = \text{ran}(c),$$

satisfies $H_a + 1 = H_a$. Consequently:

1. H_a is dense in $\widehat{\mathbb{Z}}$, and its projection onto every $\mathbb{Z}/n\mathbb{Z}$ is surjective.
2. $\mathcal{W}$ cannot be finite.
3. If $\mathcal{W}$ is compact in the Stone topology on ultrafilters on q , then $H_a = \widehat{\mathbb{Z}}$ and $|\mathcal{W}| \geq 2^{2^{\aleph_0}}$.

Proof. The space $\widehat{\mathbb{Z}}$ and its subsets are real-coded low-rank sets. By Lemma 2.2, H_a is a ground object; more precisely it is an element of the ground set $(\mathcal{P}(\widehat{\mathbb{Z}}))^M$. Its definition uses c through a , and the fixed definition of $\mathcal{W}$ from ground parameters, q , and ordinals.

For a late deletion d of c , put $a' = \text{ran}(d)$. The family $\mathcal{W}$ is unchanged in the same extension. By (7.3), $H_{a'} = H_a + 1$. Applying Lemma 3.1 to the phase-image observation gives $H_{a'} = H_a$, hence $H_a + 1 = H_a$.

Choose $z \in H_a$. The equality implies $z + \mathbb{Z} \subseteq H_a$. This orbit is dense in $\widehat{\mathbb{Z}}$: a basic nonempty open set specifies finitely many compatible congruences, which can be combined into one congruence modulo the least common multiple of their moduli. An ordinary integer translate of z satisfies that congruence. In particular each residue modulo n occurs. Since this is true for every n , the image and hence $\mathcal{W}$ cannot be finite.

For the compactness assertion, the Stone topology has basic clopen sets $\{W : B \in W\}$ for $B \subseteq q$. Each coordinate map $W \mapsto r_n^a(W)$ is continuous because its fibres are precisely such clopen sets, with $B = E_{n,r}^a$. Thus t_a is continuous. The image of compact $\mathcal{W}$ in the Hausdorff space $\widehat{\mathbb{Z}}$ is compact and closed. Being also dense, it is all of $\widehat{\mathbb{Z}}$.

Finally $|\widehat{\mathbb{Z}}| = 2^{\aleph_0}$. A compatible residue sequence is a real code, giving the upper bound. For the lower bound, arbitrary binary digits determine a 2-adic component; assign zero residues at every odd prime power, and use the Chinese remainder theorem to obtain a compatible sequence at all moduli. This injects 2^ω into $\widehat{\mathbb{Z}}$. This explains the first cardinal bound obtainable from the phase map. For the sharper bound on the family itself, apply Lemma 7.5 below: every infinite compact family of ultrafilters on a full power set has at least $2^{2^{\aleph_0}}$ members. The family here is infinite by (2). $\square$

Corollary 7.2 (Ultrafilter-valued kernels). There is no nonempty finite $\text{OD}_{M \cup \{q\}}^V$ family of functions that assign to every $x \in Q_\kappa$ an ultrafilter on $\mathcal{P}(x)^V$.

Proof. Evaluate every function in the family at q . The resulting set is a nonempty finite family of ultrafilters on q , definable from the same parameters and q , contrary to Theorem 7.1. $\square$

7.2 Sharpness: a countable local family exists

The density conclusion is compatible with countability, and in this case that compatibility has an explicit realization. It is important to distinguish a family of ultrafilters on one class from a family of global kernels on all classes.

Theorem 7.3 (The exact local threshold is countable). Fix a nonprincipal ultrafilter D on ω in M . There is a countably infinite family $\mathcal{U}_D(q)$ of nonprincipal ultrafilters on $\mathcal{P}(q)^V$ definable from D, q . Its phase image relative to $a = \text{ran}(c)$ is exactly

$$\{t_a(W) : W \in \mathcal{U}_D(q)\} = -\rho(D) + \mathbb{Z} \subseteq \widehat{\mathbb{Z}},$$

where $\rho(D)$ records the residues modulo n chosen by D on ω . The phase map is injective on this family. Consequently, the least possible cardinality of a nonempty $\text{OD}_{V_\kappa \cup \{q\}}^V$ family of ultrafilters on q is exactly $\aleph_0$. The displayed family is not compact.

Proof. There are no new subsets of ω , so the ground ultrafilter D is an ultrafilter on the full $\mathcal{P}(\omega)^V$. For $b \in q$, write $b^{(k)}$ for the representative obtained by deleting the first k elements of b , and define the pushforward ultrafilter

$$W_D^b = \{B \subseteq q : \{k < \omega : b^{(k)} \in B\} \in D\}.$$

Preimages under $k \mapsto b^{(k)}$ preserve complements and finite intersections, so this is an ultrafilter. It is nonprincipal because the representatives $b^{(k)}$ are distinct and the preimage of a singleton has at most one element. Put

$$\mathcal{U}_D(q) = \{W_D^b : b \in q\}.$$

This set is definable from q, D without choosing a representative of q . We now prove that, despite its apparently κ -sized indexing set, it has exactly countably many distinct members.

Fix any $a \in q$ for the proof. Suppose $b, b' \in q$ satisfy $\chi_a(b) = \chi_a(b')$. Choose a bound above the finite symmetric difference of b and b' . Beyond that bound they have a common tail. The numbers of their elements before that tail are equal: their difference is $\chi_b(b') = \chi_a(b') - \chi_a(b) = 0$. Thus

$$b^{(k)} = (b')^{(k)} \quad \text{for all sufficiently large } k.$$

For every $B \subseteq q$ the two preimages used to define W_D^b and $W_D^{b'}$ differ finitely, and D ignores finite differences. Therefore $W_D^b = W_D^{b'}$. There is at most one ultrafilter in the family for each integer value of $\chi_a(b)$.

For the converse and the phase computation, let $\rho_n(D)$ be the unique residue $s \pmod{n}$ for which $\{k : k \equiv s \pmod{n}\} \in D$. These residues are compatible, defining $\rho(D) \in \widehat{\mathbb{Z}}$. The cocycle identity gives

$$\chi_a(b^{(k)}) = \chi_a(b) - k.$$

It follows directly from the pushforward definition that

$$r_n^a(W_D^b) = \chi_a(b) - \rho_n(D) \pmod{n}, \quad t_a(W_D^b) = \chi_a(b) - \rho(D).$$

Distinct integer indices give distinct elements of $\widehat{\mathbb{Z}}$: a nonzero integer is nonzero modulo some positive integer. Thus they give distinct ultrafilters. Every integer index occurs, because one can delete any finite number of points from a or add any finite number of new points to it. There are enough new points since κ is uncountable and a is countable. Hence the family is in bijection with $\mathbb{Z}$, and its phase map has exactly the claimed image.

The parameter D belongs to V_κ . The construction therefore gives a nonempty countable family in $\text{OD}_{V_\kappa \cup \{q\}}^V$, while Theorem 7.1 excludes every nonempty finite family in that class. This proves the exact threshold. Its countable phase image is proper and dense in the uncountable Hausdorff space $\widehat{\mathbb{Z}}$, so it is not closed. Continuity of the phase map rules out compactness of the family. $\square$

Remark 7.4 (Local families versus global kernels). The construction defines a single assignment $x \mapsto \mathcal{U}_D(x)$ on Q_κ whose values are countable sets of ultrafilters. It does not provide a countable definable family of single-valued kernels $x \mapsto W_x$. Selecting or aligning a labeling of the local families for all x is an additional problem. Thus the local countable question is answered positively, but the archive's question about a countable family of global ultrafilter-valued kernels remains unsettled here. In particular a definable countable local family need not have a definable enumeration: such an enumeration would yield a definable first member, contradicting Theorem 7.1.

7.3 The compact threshold is also attained

The stronger compactness bound uses the full ambient Boolean algebra $\mathcal{P}(q)$. The following elementary fact is independent of Prikry forcing. Its proof also fixes the exact cardinal needed in the conclusion.

Lemma 7.5 (Infinite compact Stone families). For any set X , every infinite compact subspace F of the space of ultrafilters on $\mathcal{P}(X)$ has cardinality at least $2^{2^{\aleph_0}}$. The space of ultrafilters on a countably infinite set has cardinality exactly $2^{2^{\aleph_0}}$.

Proof. Write $[A] = \{W : A \in W\}$ for the basic clopen set associated with $A \subseteq X$. We first construct pairwise disjoint sets $A_n \subseteq X$ and points $W_n \in F \cap [A_n]$. Begin with $B_0 = X$ and an infinite trace $F \cap [B_0]$. Given B_n with infinite trace, separate two distinct ultrafilters in that trace by some subset of X . This partitions B_n into two pieces whose traces in F are both nonempty. At least one trace is infinite. Retain that piece as B_{n+1} and call the other piece A_n . Choose W_n in its nonempty trace. The A_n are pairwise disjoint by construction.

For any ultrafilter D' on ω , define

$$W_{D'} = \{A \subseteq X : \{n : A \in W_n\} \in D'\}.$$

It is an ultrafilter because this coordinatewise test preserves Boolean operations. It lies in the closure of $\{W_n : n < \omega\}$: if $A \in W_{D'}$, then some W_n contains A , and finite intersections of such basic neighborhoods reduce to a single basic neighborhood. The compact subspace F is closed in the Hausdorff Stone space, so $W_{D'} \in F$.

For $S \subseteq \omega$, put $C_S = \bigcup_{n \in S} A_n$. Disjointness gives $C_S \in W_n$ exactly when $n \in S$. Therefore

$$C_S \in W_{D'} \iff S \in D'.$$

Distinct ultrafilters D' produce distinct points of F . Thus F contains at least as many points as there are ultrafilters on ω .

Here is a proof of the latter cardinality. An ultrafilter is a subset of $\mathcal{P}(\omega)$, so the upper bound is $2^{2^{\aleph_0}}$. For the lower bound, use the countable set

$$Z = \{(n, H) : n < \omega, H \subseteq 2^n\}.$$

For each binary sequence $x \in 2^\omega$, let $A_x = \{(n, H) : x \upharpoonright n \in H\}$. Given finitely many distinct $x_1, \dots, x_k$ and prescribed choices of A_{x_i} or their complements, their corresponding prefixes are distinct for all sufficiently large n . At every such n one can choose H to meet exactly the prescribed membership requirements. Thus every such finite Boolean combination is infinite. For each of the $2^{2^{\aleph_0}}$ assignments of a zero or one to all the indices $x \in 2^\omega$, the chosen sides, together with the cofinite filter on Z , have the finite intersection property and extend to an ultrafilter. Different assignments require different ultrafilters. Since Z is countably infinite, this proves the lower bound on ω as well. $\square$

Corollary 7.6 (The exact compact local threshold). The least cardinality of a nonempty compact $\text{OD}_{V_\kappa \cup \{q\}}^V$ family of ultrafilters on q is exactly $2^{2^{\aleph_0}}$. A family attaining this bound is the Stone closure

$$C_D(q) = \overline{\mathcal{U}_D(q)}.$$

All its members are nonprincipal, and its phase image is all of $\widehat{\mathbb{Z}}$.

Proof. The Stone space of ultrafilters on $\mathcal{P}(q)$ is compact Hausdorff: it is the closed space of Boolean homomorphisms to 2 inside $2^{\mathcal{P}(q)}$. Thus the displayed closure is compact. It is nonempty and definable from D, q ; explicitly, membership in the closure says that every basic neighborhood meets $\mathcal{U}_D(q)$. It is infinite since it contains that countably infinite family, so Lemma 7.5 gives the lower bound.

For the matching upper bound, fix $a \in q$ for the proof and let $R_a = \{a^{(k)} : k < \omega\}$. Every $b \in q$ has a tail ray that merges with R_a , so $R_a \in W_D^b$. Hence the whole countable family lies in the clopen subspace $[R_a]$, and so does its closure. Restriction to R_a identifies $[R_a]$ with the ultrafilters on the countably infinite set R_a . This space has cardinality $2^{2^{\aleph_0}}$ by Lemma 7.5, proving the upper bound.

Every principal ultrafilter is an isolated point, and the set of all principal ultrafilters is open. Therefore the nonprincipal ultrafilters form a closed subspace. The family in Theorem 7.3 lies there, and so does its closure. The phase image of the closure is all of $\widehat{\mathbb{Z}}$ by Theorem 7.1. Finally any nonempty compact definable family is infinite by that same theorem, and hence has at least the displayed cardinality by Lemma 7.5. This proves minimality among all such definable families, not only for the example. $\square$

The positive constructions in Theorem 7.3 and Corollary 7.6 do not intrinsically require Prikry forcing: they work for every class in $\mathcal{Q}_\lambda$, for any uncountable λ of cofinality ω , using a nonprincipal ultrafilter on the ambient $\mathcal{P}(\omega)$. The Prikry hypothesis supplies the nonexistence of smaller definable families at the generic class.

8 Charges: the exact finite-additivity contrast

A *charge* on q is a finitely additive probability measure $\mu : \mathcal{P}(q)^V \rightarrow [0, 1]$. Countable additivity is not included in the definition. We first describe a restriction on definable finite families and then construct examples showing why the restriction is not stronger.

Theorem 8.1 (Residue laws for finite families of charges). Suppose $\mathcal{M} \in \text{OD}_{M \cup \{q\}}^V$ is a nonempty family of charges on q and $h = |\mathcal{M}| < \omega$. Let $a = \text{ran}(c)$.

1. If $h = 1$, its member μ satisfies $\mu(E_{n,r}^a) = 1/n$ for all $n \geq 1$ and every residue r .
2. For general h , every $\mu \in \mathcal{M}$ satisfies $\mu(E_{\ell,r}^a) = 1/\ell$ for every prime $\ell > h$ and every residue r .
3. Every member has infinitely many values, gives measure zero to each integer-index fibre $\chi_a^{-1}\{k\}$, and fails countable additivity.

Proof. For each modulus n , form the set of probability vectors

$$H_n(a) = \{(\mu(E_{n,r}^a))_{r < n} : \mu \in \mathcal{M}\} \subseteq [0, 1]^n.$$

It is nonempty, has size at most h , and belongs to M : each entry is a real, there are no new reals, and the set of vectors is finite. On deleting one point of a , the r th coordinate becomes the old $(r - 1)$ st coordinate. The deletion lemma makes $H_n(a)$ invariant under this cyclic rotation.

When $h = 1$, the one vector is fixed by the rotation, so all its coordinates are equal. Their sum is 1 by finite additivity, hence they are all $1/n$. For a prime modulus $\ell > h$, an orbit of a vector under rotation has length either 1 or ℓ . Since the invariant set of vectors has at most $h < \ell$ elements, every vector is fixed. Again its coordinates must all be $1/\ell$.

For a fixed $\mu \in \mathcal{M}$, the values $1/\ell$ for primes $\ell > h$ are infinitely many distinct members of its range. For each integer k ,

$$\chi_a^{-1}\{k\} \subseteq E_{\ell, k \bmod \ell}^a, \quad 0 \leq \mu(\chi_a^{-1}\{k\}) \leq 1/\ell.$$

Letting these primes grow shows that this fibre has measure zero. The integer-index fibres form a countable partition of q . Countable additivity would therefore give $\mu(q) = 0$, contradicting $\mu(q) = 1$. $\square$

In particular no nonempty finite definable family can consist of finite-range charges, or of countably additive probability measures. Ultrafilters correspond to the two-valued charges and are an immediate special case, although Theorem 7.1 says substantially more about arbitrary ultrafilter families.

8.1 An explicit positive kernel

The supplied synthesis already proves the existence of a definable charge-valued kernel from a shift-invariant mean. The following is a concrete Cesàro–ultrafilter implementation of that positive construction, not a new claim of existence.

Fix a nonprincipal ultrafilter D on ω in M . Such a D exists by Choice, lies in V_κ , and remains an ultrafilter on all subsets of ω in V because no new reals are added. For $a \in D_\kappa$, let $a^{(k)}$ be a with its first k elements deleted. If $x = [a]$ and $B \subseteq x$, define

$$K_D(x)(B) = \lim_{N \rightarrow D} \frac{1}{N} |\{k < N : a^{(k)} \in B\}|. \quad (8.1)$$

The ultrafilter limit of a bounded real sequence is the unique limit along D in the compact interval $[0, 1]$.

Proposition 8.2 (A definable finitely additive kernel). Formula (8.1) is independent of the chosen representative $a \in x$. It defines a charge on every $x \in Q_\kappa$, and the entire kernel K_D is ordinal definable from D . In particular, finitely additive probability kernels exist in $\text{OD}_{V_\kappa}^V$.

Proof. If $a =^* b$, choose $r, s < \omega$ such that $a^{(r)} = b^{(s)}$. Then $a^{(r+j)} = b^{(s+j)}$ for every $j < \omega$. For each $B \subseteq [a]$, the two sequences of indicator values along these rays have a common tail after shifting their indices by a fixed integer. Their first N averages differ by at most a constant divided by N . This tends to zero along the nonprincipal ultrafilter D , so the limits agree.

For disjoint B, C , the indicator of their union is the sum of the two indicators at every point of the ray. Taking finite averages and then the ultrafilter limit preserves this equality. The empty set has value 0 and the whole class has value 1, while all values are nonnegative. Thus the formula defines a charge. Independence permits the definition to quantify over a representative without making a choice of one; the result is a single function of x and D . Replacement gives the kernel as a set, and its definition uses only D and the ordinal κ . $\square$

This proof uses no embedding hypothesis. The ray-average method works for any uncountable λ of cofinality ω in a universe where the chosen D is an ultrafilter on its full $\mathcal{P}(\omega)$. At the generic class it has the uniform residue values required by Theorem 8.1. It also assigns zero to each singleton: the ray $a^{(0)}, a^{(1)}, \dots$ has distinct terms, so a singleton can occur at most once in an average.

8.2 Sharpness for finite families

It would be incorrect to infer from Theorem 8.1 that every member of a finite definable family is uniform at *every* modulus. There are exact finite examples showing the difference between a singleton and an unordered family.

Proposition 8.3 (Periodic ray averages). For every integer $d \geq 1$ there is a d -element family of charges on q definable from q, D . When $a = \text{ran}(c)$ is used to label its members as $\mu_0^a, \dots, \mu_{d-1}^a$, one has

$$\mu_r^a(E_{d,-r}^a) = 1.$$

For a prime d , this shows that the threshold $\ell > h$ in Theorem 8.1(2) cannot be replaced by $\ell \geq h$.

Proof. For any $a \in q$ put

$$\mu_r^a(B) = \lim_{N \rightarrow D} \frac{1}{N} |\{j < N : a^{(dj+r)} \in B\}|, \quad 0 \leq r < d.$$

Each is a charge by the same finite-additivity argument as above. If $a^{(r_0)} = b^{(s_0)}$, then their full tail rays agree after the fixed shift $s_0 - r_0$. The subray with indices congruent to r modulo d therefore agrees, after a finite shift of its own index, with the subray of b having residue $r + s_0 - r_0$ modulo d . Finite shifts do not affect the averages. Thus changing a permutes the set $\{\mu_r^a : r < d\}$, and that unordered set is definable from q, D .

Since $\chi_a(a^{(dj+r)}) = -(dj+r)$, every term in this subray belongs to $E_{d,-r}^a$. The corresponding charge assigns this cell value 1 and every other cell of the d th partition value 0. The d charges are therefore distinct. For prime $d \geq 2$, none is uniform at modulus d , even though the definable family has exactly d members. $\square$

The last construction is a finite family of *local charges on one class*. It does not choose a labeling of these charges simultaneously across all classes and therefore should not be silently reinterpreted as a finite family of global kernels. The singleton kernel in Proposition 8.2 has no such ambiguity.

9 The ground trace, rigidity, and the canonical core

The measurable-strength rigidity and normal-trace constructions are already in the supplied synthesis [1, Section 13]. Here we prove the specific version needed to put the new finite-choice conclusions in the *same* model. This section also shows that those conclusions do not force the normal trace to concentrate on smaller measurables.

9.1 Homogeneity relative to the generic class

Lemma 9.1 (Ground containment for definable sets of ordinals). Every set of ordinals in $\text{OD}_{M \cup \{q\}}^V$ belongs to M .

Proof. First consider a sentence with ground parameters, ordinal parameters, and the canonical name for the generic finite-difference class. Any two conditions can be strengthened to conditions whose cones are isomorphic by replacing the stems.

Explicitly, given (s, A) and (t, B) , shrink the upper parts to a common $C \in U$ contained in $A \cap B$ and above both stems. Below (s, C) , conditions have stems s followed by a finite sequence u from

C. Replace that stem by t followed by the same u , leaving the later upper part unchanged. This is an isomorphism with the cone below (t, C) . Corresponding generic presentations have the form s followed by a common infinite tail and t followed by that tail. They generate the same extension and have the same finite-difference class. Ground parameters are interpreted identically. Consequently the original sentence has the same truth value in the corresponding presentations.

This is the usual forcing-invariance argument for isomorphic cones, with invariance of the class parameter checked explicitly. In particular, two conditions cannot force opposite truth values for such a sentence. Since deciding conditions are dense, the maximum condition decides every such sentence.

Now let $x \subseteq \eta$ be uniquely definable in V from allowed parameters. The assertion that there is a unique such subset of η is true in V and is therefore forced by the maximum condition, by the preceding paragraph. For each $\alpha < \eta$, the maximum condition also decides whether α belongs to that uniquely defined set. Thus

$$x = \{\alpha < \eta : \mathbb{1}_{\mathbb{P}_U} \Vdash \alpha \text{ belongs to the uniquely defined subset}\}.$$

The forcing relation for this fixed formula and its ground parameters is definable in M . Separation in M therefore puts x in M . Any set of ordinals is bounded by some ordinal, so the claim follows. $\square$

Lemma 9.2 (Hereditary ground containment). Both

$$N = \text{HOD}_{V_\kappa \cup \{q\}}^V \quad \text{and} \quad H_p = \text{HOD}_{\{p, q\}}^V \quad (p \in M)$$

are subclasses of M .

Proof. Use induction on rank in either hereditary class. Suppose x belongs to that class and all its elements belong to M by induction. Choose an ordinal β with $x \subseteq V_\beta^M$, and choose in M an ordinal-indexed enumeration $e : \theta \rightarrow V_\beta^M$. Such an enumeration exists by Choice in M . The set

$$I_x = \{\xi < \theta : e(\xi) \in x\}$$

is a set of ordinals definable from q and ground parameters. Here the parameters formerly allowed from V_κ^V belong to M by Lemma 2.2; the additional parameter e is also in M . Lemma 9.1 puts I_x in M . Hence $x = e[I_x] \in M$. The empty-rank case starts the induction. $\square$

9.2 Regularity and rigidity

The class N above is the hereditary definability inner model used in the synthesis; it satisfies ZF and contains V_κ . The latter inclusion follows directly because every element of V_κ , and every member of its transitive closure, is an allowed parameter. Choice is not needed in N for the following regularity argument.

Proposition 9.3 (A rigid generic class). The cardinal κ is regular in N . Equivalently in the terminology of the synthesis, q is rigid: no nonempty family of fewer than κ cofinal subsets of κ of order type below κ belongs to $\text{OD}_{V_\kappa \cup \{q\}}^V$. Also, for $T \in \text{OD}_{V_\kappa \cup \{q\}}^V$ with $T \subseteq D_\kappa$,

$$|T \cap q| \in \{0, \kappa\}.$$

Proof. A cofinal map from an ordinal below κ to κ belonging to N would belong to M by Lemma 9.2. It would still be cofinal there, contradicting regularity of the measurable cardinal κ in M . Thus κ is regular in N .

Suppose $\mathcal{A} \in \text{OD}_{V_\kappa \cup \{q\}}^V$ is a nonempty family of the indicated kind with $|\mathcal{A}| < \kappa$. Let μ be the least order type occurring among its members, and let $\mathcal{A}_\mu$ consist of the members of that order type. Their union u is cofinal in κ and is definable from the allowed parameters. Moreover

$$|u| \leq |\mathcal{A}_\mu| \cdot |\mu| < \kappa.$$

This inequality uses only a product of two fixed cardinals below the infinite cardinal κ , not an invalid regularity assumption about κ in V . Because $u \subseteq \kappa$, its order type is below κ . As a definable set of ordinals, u belongs to N , and its increasing enumeration there contradicts regularity. Conversely a short cofinal subset in N would give a forbidden singleton family; this also explains the equivalence with the archive's rigidity formulation.

Every class q has cardinality κ . Fix $a \in q$. Every member is obtained from a by finitely many additions and deletions, giving the upper bound κ . Adding one point $\alpha \in \kappa \setminus a$ gives κ distinct members, giving the lower bound. If $T \cap q$ were nonempty of size below κ , it would be a forbidden definable family of cofinal ω -sets. This proves the last statement. $\square$

9.3 The internal normal measure

Choose in M a bijection $b : \kappa \rightarrow V_\kappa^M$, using the inaccessibility of κ there, and set

$$H = \text{HOD}_{\{b, q\}}^V, \quad U_H = \{A \in \mathcal{P}(\kappa) \cap H : \text{Tail}(q, A)\}.$$

Unlike the many-parameter model N , the hereditary model with the two fixed set parameters b, q satisfies ZFC: the usual definition-code well-order of ordinal-definable objects gives Choice in this hereditary class. We use that standard interpretation of $\text{HOD}_{\{b, q\}}^V$ throughout.

Theorem 9.4 (A normal tail measure in the canonical core). The following hold:

1. $H \subseteq M$, $V_\kappa^H = V_\kappa^V$, and $b \in H$.
2. $U_H \in H$, and H regards it as a normal nonprincipal κ -complete ultrafilter on κ . Externally, $U_H = U \cap H$.
3. Every representative of q is Prikry generic over H for U_H ; all give the same extension $H[c]$, with

$$H \subsetneq H[c] \subseteq V, \quad V_\kappa^{H[c]} = V_\kappa^V.$$

The extension $H[c]$ has the same cardinals as H , contains q , and has cofinality ω at κ . No representative of q belongs to H .

Proof. Lemma 9.2 gives $H \subseteq M$. For every $x \in V_\kappa^M$, choose $\alpha < \kappa$ with $x = b(\alpha)$. This makes x definable from b and an ordinal. The same is true of every member of its transitive closure, because all those members also lie in V_κ^M . Thus $x \in H$. Together with Lemma 2.2 this gives equality of the rank segments. The graph b is itself definable from b , and its elements and their descendants consist of ordinals and these same low-rank sets. Hence $b \in H$.

For $A \in \mathcal{P}(\kappa) \cap H$, we have $A \in M$. If $A \in U$, Mathias' criterion gives $a_c \subseteq^* A$. If $A \notin U$, its complement belongs to U , so a_c is almost disjoint from A and cannot be almost contained in A . Therefore

$$\text{Tail}(q, A) \iff A \in U,$$

which proves $U_H = U \cap H$. The definition of U_H from b, q makes it ordinal definable from those parameters, and every one of its members is in H . Therefore it is hereditarily so definable, and $U_H \in H$. It is an ultrafilter on the subsets of κ in H , and is nonprincipal, by the corresponding properties of U .

For internal completeness, take a sequence $\langle A_i : i < \delta \rangle \in H$, where $\delta < \kappa$ and all $A_i \in U_H$. The entire sequence belongs to M . Completeness of U in M puts its intersection in U ; that intersection belongs to H , so it belongs to U_H . For normality, take in H a regressive function on a set in U_H . This is a ground regressive function on a member of U . Normality in M provides a constant fibre in U . The fibre is defined from the function and its ordinal value, so it belongs to H , hence to U_H . These are precisely the internal completeness and normality assertions.

Every $a \in q$ is cofinal of order type ω and is almost contained in every $A \in U_H$, by definition. Mathias' criterion, now over H , proves its genericity. Finite changes yield the same extension because the finitely many extra ordinals belong to H . Prikry preservation over H gives equality of cardinals and the new cofinality. Every finite modification of a can be formed in $H[a]$; conversely every member of q is such a modification. Therefore its internally computed class is exactly the ambient q , so $q \in H[a]$. Since H already contains the full ambient V_κ , that rank segment stays equal in the intermediate model. Finally H sees κ measurable and hence regular; a representative of q would give it a cofinal ω -sequence. Thus no representative belongs to H , and the inclusion $H \subsetneq H[c]$ is proper. $\square$

Two different ultrafilters, with different domains

The internal normal measure U_H is an ultrafilter on $\mathcal{P}(\kappa) \cap H$, and its completeness is tested on sequences belonging to H . The forbidden ultrafilters of Theorem 7.1 live on $\mathcal{P}(q)^V$. These statements do not conflict. In fact the ambient countable sequence of tail intervals of c demonstrates that internal κ -completeness of U_H is not ambient countable completeness.

Corollary 9.5 (The new obstructions do not force concentration). Suppose κ is the least measurable cardinal of M . Then H regards κ as its least measurable cardinal, while all the finite-family conclusions above hold in V . In particular U_H does not concentrate on the cardinals below κ that are measurable in H .

Proof. For $\alpha < \kappa$, the objects needed to express that α carries a nonprincipal α -complete ultrafilter all lie in V_κ : subsets of α , candidate ultrafilters on them, and sequences of length below α of their elements have rank bounded below the inaccessible ordinal κ . The models H and M agree on this entire rank segment. They therefore agree about measurability below κ . There are no such measurables in M , so there are none in H . Theorem 9.4 makes κ measurable in H . The set of internally measurable cardinals below it is empty and cannot be in U_H . All earlier arguments required only that U be a normal measure on κ , so they still apply. $\square$

This is a separation *at the displayed witness*: the new finite choice failures do not recover the concentration-on-measurables feature of the ultraextending critical-sequence construction in the archive. It is not a claim that no stronger normal-trace witness exists anywhere else in the universe.

10 The precise consistency calibration

We give a statement without the external ground-model symbol M . This prevents a conditional model construction from being confused with an unqualified new lower-bound theorem for bare selector failure.

Definition 10.1 (The finite cyclic obstruction at a class). For a strong limit uncountable cardinal λ of cofinality ω and $q \in Q_\lambda$, let $\text{FC}(\lambda, q)$ assert the following. For every finite $\Gamma \leq S_m$ and every nonempty

finite Γ -set Y coded by natural numbers, if $\text{CF}(\Gamma, Y)$ fails, then there is no nonempty finite family of admissible rules $D_m(\lambda) \rightarrow Y$ in $\text{OD}_{V_\lambda \cup \{q\}}$.

The sufficiency half of the finite-label classification needs no extra axiom here: the incidence-word construction uses a well-order of the reals, which is an available parameter in V_λ . Thus $\text{FC}(\lambda, q)$ includes the full equivalence between the cyclic condition, existence of a definable rule, and existence of a nonempty finite definable family. It also implies every finite-selector obstruction in Corollary 5.4.

Definition 10.2 (The calibrated package). Let $\text{NT} + \text{FC}$ assert that there exist λ, b, q such that:

1. λ is an uncountable strong limit cardinal with $\text{cf}(\lambda) = \omega$, $q \in Q_\lambda$, and $b : \lambda \rightarrow V_\lambda$ is a bijection;
2. λ is regular in $\text{HOD}_{V_\lambda \cup \{q\}}$;
3. in $H = \text{HOD}_{\{b, q\}}$, the set $\{A \in \mathcal{P}(\lambda) \cap H : \text{Tail}(q, A)\}$ belongs to H and is a normal nonprincipal λ -complete ultrafilter on λ ;
4. $\text{FC}(\lambda, q)$ holds.

These are assertions about definable inner classes in the usual language of set theory. Relative ordinal definability and membership in the hereditary classes have their standard uniform first-order definitions; no satisfaction predicate for the whole universe is being added.

Theorem 10.3 (Exact measurable-strength package). Over ZFC,

$$\text{Con}(\text{ZFC} + \text{there is a measurable cardinal}) \iff \text{Con}(\text{ZFC} + \text{NT} + \text{FC}).$$

The equivalence remains true if $\text{NT} + \text{FC}$ also requires that the displayed λ be the *least* measurable cardinal of its displayed core $\text{HOD}_{\{b, q\}}$.

Proof. For the upper bound, start with a measurable cardinal and a normal measure in a ground model of ZFC. Force with its ordinary Prikry forcing. Lemma 2.2 supplies the strong limit singular cardinal and the ground bijection $b : \kappa \rightarrow V_\kappa$. Proposition 9.3 supplies regularity in the many-parameter hereditary model. Theorem 9.4 supplies the normal tail measure in the fixed-parameter core. Finally Theorem 5.2, using $\text{OD}_{V_\kappa \cup \{q\}}^V \subseteq \text{OD}_{M \cup \{q\}}^V$, supplies $\text{FC}(\kappa, q)$. Therefore the extension satisfies $\text{NT} + \text{FC}$. Choosing the least measurable cardinal in the ground and applying Corollary 9.5 gives the additional clause.

For the lower bound, any model of the package has the definable inner model $H = \text{HOD}_{\{b, q\}}$ satisfying ZFC. Clause (3) says exactly that H has a measurable cardinal, witnessed by the displayed internal ultrafilter. Thus it yields a model of the measurable-cardinal theory.

These are the standard relative-consistency arguments by set forcing and by interpretation in a definable inner model. The transitive-model notation used to present the mathematics is not an extra hypothesis on the two consistency statements. Equivalently one can apply the forcing interpretation and the inner-model interpretation to each finite proof that might witness inconsistency. $\square$

Remark 10.4 (Where the lower bound comes from). The lower bound in Theorem 10.3 comes from the normal-trace clause, not from $\text{FC}(\lambda, q)$ alone. The archive already calibrated normal traces at measurable strength. The contribution is that adding all the formerly ultraexacting finite cyclic obstructions does not increase that strength. No theorem here identifies the exact consistency strength of the bare failure of definable binary choice on a cofinal quotient.

The upper-bound model also has the profinite-density theorem and the finite-family charge restrictions from Sections 7 and 8. Those are consequences of the forcing construction; they need not be included in the definition of the package to obtain its stated calibration.

11 Consequences for the research program

A resolved forcing question. The Prikry-extension part of the synthesis's last question group (ii) is settled by Theorems 5.2 and 6.2 and Corollary 5.4. The result is uniform in every ordinary normal-measure Prikry extension, not just in a specially selected model. The allowed definition parameters include the invariant generic class q and arbitrary ground sets. This is stronger than merely removing ultraexactness from the OD_{V_κ} singleton conclusion.

A distinction between two parts of the ultraexact picture. Finite cyclic failures and the existence of one normal-trace core can coexist when that core has no smaller measurable cardinal. They therefore do not reproduce, at the same witness, the concentration-on-measurables property established for the ultraexact critical-sequence trace in the archive. The latter remains a genuinely additional property in this comparison. Nothing here transfers the archive's internal successor measurability theorem to the present Prikry construction.

A structural obstruction rather than only a cardinal bound. For ultrafilters, the conclusion is the exact dynamical relation $H_a + 1 = H_a$ on a canonically extracted phase image. Density, all finite residue surjections, and the compactness lower bound follow from it. Theorem 7.3 realizes a single dense integer orbit and shows that the least size of a nonempty definable local family is exactly $\aleph_0$. Corollary 7.6 gives the exact compact threshold $2^{2^{\aleph_0}}$, also with a definable example. For charges, finite families really can exist, and the periodic ray examples exhibit the sharp obstruction at prime moduli. Thus replacing ultrafilters by finitely additive measures changes the answer in a way that is both constructive and quantitatively visible.

Question 11.1 (Rigidity without a Prikry presentation). Does regularity of λ in $\text{HOD}_{V_\lambda \cup \{q\}}$, by itself, imply $\text{FC}(\lambda, q)$? The proof here needs a late-deletion principle for ground-valued observations. No derivation of that principle from regularity alone is supplied.

Question 11.2 (Countable global families). Can a countably infinite OD_{V_κ} family of single-valued ultrafilter-valued kernels on Q_κ exist? The countable local families of Theorem 7.3 do not furnish a countable family of global sections. The analogous question for countable families of selectors also remains outside the present proof.

Question 11.3 (A wider forcing criterion). Which other singularizing forcings admit both a same-extension deletion operation and enough ground-valued response tables to reproduce Theorem 5.2? An application must verify those properties for the actual forcing; no blanket assertion is made here for all Prikry-type or tree-Prikry forcings.

A Finite checks and a proof audit

The archive accompanying this report contains `finite_check.py`, a dependency-free Python program, and its actual output in `finite_check_results.txt`. The program checks four elementary parts of the construction: the root identity in Lemma 4.2; the sign and index in the deletion formula using finite integer sequences with disjoint blocks; the absence of a fixed proper subset under a full cycle; and the S_3 example with elementwise but no simultaneous fixed points. It also checks the finite residue frequencies behind the periodic ray averages.

These tests are useful for catching sign errors in g^{-1} or τ^{-n} . They are not tests of a generic extension, of the forcing theorem, of hereditary ordinal definability, or of any large-cardinal consistency assertion. No Lean formalization of the new arguments is included, and none is claimed.

Potential gap	How it is avoided
Replacing a definable finite family by a definable member	The proof uses the unordered set of all finite response tables.
Pretending that deletion is an automorphism of V	The forcing truth lemma compares two generic presentations of the same universe.
Using a new high-rank response as a ground constant	Only finite tables with ground entries, or genuinely unchanged low-rank objects, are decided.
Confusing the generic class with a representative	The class q is invariant; c is used only in the observation and in test configurations.
Using regularity at the ambient singular κ	Small unions are bounded using two fixed cardinal factors; regularity is used only in the inner model.
Equating internal and ambient measures	U_H is on internal subsets of κ ; forbidden ultrafilters are on full ambient subsets of q .
Overstating the consistency result	Its lower bound is explicitly supplied by the normal-trace clause.
Confusing the two size thresholds	The local minimum is $\aleph_0$; the compact minimum is $2^{2^{\aleph_0}}$, using a separate Stone-space argument.

B Source and notation concordance

The notation $D_\lambda, Q_\lambda, \text{Tail}, \text{CF}$ follows the supplied *Synthesis*. Its “rigidity is regularity” theorem is used here in the explicit small-family form of Proposition 9.3. Its fixed-parameter normal-trace model $\text{HOD}_{\{b,q\}}$ is denoted by H here; M is reserved for the actual Prikry ground. Our $V = M[c]$ is the ambient extension, not the internal Prikry core $H[c]$.

The synthesis’s finite-label sufficiency proof is expanded in Section 6. Its measurable-strength trace construction is expanded in Section 9. Neither reproduction is offered as a new theorem. The new forcing transfer is Theorem 5.2; the main further refinements are Theorems 7.1, 7.3, and 8.1, Corollary 7.6, Proposition 8.3, and the compatibility/separation statements in Corollary 9.5 and Theorem 10.3.

The supplied Lean development includes files such as `Cardinals/FiniteLabel.lean`, `Cardinals/Combinatorics/FiniteCycles.lean`, and `Cardinals/Combinatorics/ThirdRound.lean`. Their role in this continuation is terminological and conceptual reference. The finite permutation tests in this archive are a separate, limited check and do not change the verification status of that development.

References

- [1] *Large cardinals at the inconsistency frontier: a synthesis*. User-supplied research archive `Cardinals4.zip`, `docs/Large_Cardinals_Synthesis.tex`, dated 18 September 2026; with `docs/Large_Cardinals_Unified_Report.tex` and the `Cardinals/Lean` reference development. Especially the sections “The exact finite-label classification,” “Measures hidden in rigid classes,” “Exact consistency strength,” and the final open-question group for reports 19–27. Unpublished supplied material.
- [2] Tom Benhamou. *Prikry forcing and tree Prikry forcing of various filters*. *Archive for Mathematical Logic* **58** (2019), 787–817. doi:10.1007/s00153-019-00660-3. Preprint: [arXiv:1801.04424](https://arxiv.org/abs/1801.04424); theorem numbers in this report refer to version 2, 25 September 2018.
- [3] A. R. D. Mathias. *On sequences generic in the sense of Prikry*. *Journal of the Australian Mathematical Society* **15** (1973), 409–414. The normal-measure criterion is also restated and proved as Theorem 3.17 in [2].

- [4] Juan P. Aguilera, Joan Bagaria, and Philipp Lücke. *Large cardinals, structural reflection, and the HOD Conjecture*. [arXiv:2411.11568](#). Background for the exacting/ultraexacting program; not an input to the new deletion proof.
- [5] Juan P. Aguilera, Joan Bagaria, Gabriel Goldberg, and Philipp Lücke. *Large cardinals beyond HOD*. [arXiv:2509.10254](#), version 1, 12 September 2025. Background connecting large embeddings, HOD, and Prikry constructions.
- [6] Gabriel Goldberg. *Consistency beyond the Kunen inconsistency*. Lecture notes reporting joint work with Doug Blue, Warsaw Inner Model Theory Meeting, 1 July 2026, especially Remark 2.4. [Author-hosted PDF](#).